Question	Answer	Mark
2(a)(i)	(work =) force $\times$ distance moved in the direction of the force	**B1**
2(b)(i)	$\rho = m / V$	**C1**
	$= (20/9.81)/(4/3 \times \pi \times 0.16^3)$ $= 120\,\text{kg}\,\text{m}^{-3}$	**A1**
2(b)(ii)	the pressure on the lower surface (of sphere) is greater than the pressure on the upper surface (of sphere)	**B1**
2(b)(iii)	$a = (170 - 20)/(20/9.81)$	**C1**
	$= 74\,\text{m}\,\text{s}^{-2}$	**A1**
2(b)(iv)	$D = 170 - 20$ (= 150)	**C1**
	$810 \times (0.16^2) \times v^2 = 150$	**C1**
	$v = 2.7\,\text{m}\,\text{s}^{-1}$	**A1**
2(b)(v)	$4870 = (4850 \times v)/(v - 6.30)$	**C1**
	$v = 1530\,\text{m}\,\text{s}^{-1}$	**A1**

Question	Answer	Mark